

Lab Sheet 08 – Neural Networks: Behavior & Design

Course: SE4010 – Current Trends in Software Engineering

Task 01 – Quick Concept Check

1. What is the role of weights in a neural network?
2. What does the bias do?
3. What is the purpose of an activation function?

Task 02 – Activation Functions

Given $z = -2$:

Compute:

1. $\text{ReLU}(z) = ?$
2. $\text{Leaky ReLU}(z) (\alpha = 0.01) = ?$

Short Questions:

1. Why does **ReLU output 0 for negative values**?
2. What is the **“Dying ReLU” problem**?

Task 03 – Network Design

Predict whether a student will **Pass (1) or Fail (0)** based on:

- Study hours
- Attendance

- Assignment marks

Tasks:

1. Number of input neurons = ?
2. Design a simple network:

Layer	Details
Input	
Hidden	
Output	